

CPR, or cardiopulmonary resuscitation, is an emergency life-saving procedure performed when someone's breathing or heartbeat has stopped, consisting of chest compressions often combined with artificial ventilation, or mouth-to-mouth in an effort to manually preserve and rescue breaths to maintain blood flow and oxygenation until medical help arrives. CPR is used on people in cardiac arrest to oxygenate the blood and maintain a cardiac output to keep vital organs alive. Immediate CPR can double or triple chances of survival after cardiac arrest.

120. In coronary artery bypass surgery, the blood vessel to be used as graft has to be –

- (a) Vein from the leg
- (b) Artery from the forearm
- (c) Artery from mammary region of chest
- (d) Any of these

U.P.P.C.S. (Pre) 2014

Ans. (d)

In Coronary Artery Disease (CAD), the arteries that supply blood and oxygen to the heart muscle grow hardened and narrowed. Coronary artery bypass surgery creates a new path for blood to flow to the heart. The surgeon takes a healthy piece of vein from the leg or artery from the chest or wrist. Then the surgeon attaches it to the coronary artery, just above and below the narrowed area or blockage. This allows blood to bypass (get around) the blockage.

121. In which of the following departments of a hospital will 'Cath lab' be found ?

- (a) Pathology
- (b) Surgery
- (c) Cardiology
- (d) Nuclear Medicine

U.P.P.C.S. (Pre) 2014

Ans. (c)

A catheterization laboratory or cath lab is a clinic with diagnostic imaging equipment used to visualize the arteries of the heart and the chambers of the heart and treat any stenosis or abnormality found. The cardiology cath lab area is, where all invasive cardiology investigations and treatments are undertaken.

122. The nutritional deficiency condition that needs to be given top priority for remedial action in India today is :

- (a) Scurvy
- (b) Rickets
- (c) Xerophthalmia
- (d) Pellagra

I.A.S. (Pre) 1996

Ans. (c)

Xerophthalmia, caused by severe vitamin A deficiency, is described by, pathological dryness of conjunctiva and cornea. Cells of cornea get dried, keratinized and vision power affected. That's why remedial action for Xerophthalmia should be prioritized.

123. Lathyrism is caused by excessive consumption of :

- (a) Khesari dal
- (b) Mustard oil
- (c) Polished rice
- (d) Mushrooms

I.A.S. (Pre) 1999

Ans. (a)

Excessive consumption of khesari dal (*Lathyrus sativus*) can lead to lathyrism, a neurological disorder characterized by muscle weakness, spasms, and paralysis, particularly of the lower limbs.

124. Which pulse causes lathyrism when consumed on a regular basis?

- (a) Khesari dal
- (b) Chana dal
- (c) Moong dal
- (d) None of the above

M.P. P.C.S. (Pre) 2024

Ans. (a)

See the explanation of above question.

125. Blank capsules used in dispensing are made of :

- (a) Egg-white
- (b) Gum
- (c) Starch
- (d) Gelatine

U.P.P.C.S.(Pre) 2001

Ans. (c) & (d)

The blank capsule used in dispensing are mainly made of gelatine which melts easily in the stomach, as a result medicine is reached in the body without disturbance. Blank capsules are also made from starch. Starch capsules have been shown to be a very useful alternative to hard gelatin capsules.

VII. Main Plant Diseases

Notes

- **Plant Pathology** is the study of the diseases and disorders of plants.
- Diseases can be defined as a harmful deviation from the normal functioning of the physiological processes caused by an infectious agent.
- In the case of plant diseases, the causal agent may be virus, bacteria, fungus or a parasitic agent or lack of nutrients.
- Plant pathology is also known as **phytopathology**.

Powdery mildew :

- Powdery mildew are plant diseases of worldwide occurrence which causes a powdery growth on the surface of leaves, buds, young shoots, fruits & flowers.
- Powdery mildew diseases are caused by many different species of fungi in the order Erysiphales, with **Podosphaera xanthii** (*Sphaerotheca fuliginea*) being the most reported case. *Erysiphe cichoracearum* was formerly reported to be the primary causal organism throughout most of the world.

- Hundreds of species of trees, shrubs, vines, flowers, vegetable, fruits, grasses, field crops and weeds can be affected by powdery mildew.
- The white powdery appearance is due to a large number of microscopic asexual spores (**conidia**) borne in chains.
- Sulphur dust is effective against many powdery mildew diseases.
- Other organic treatments include copper based fungicides, baking soda solutions and neem oil have also proven effective.

Green ear disease :

- The green ear disease of bajra is a common disease and has been reported from several countries including India, Iran, Israel, China, Fiji, Japan and the countries wherever bajra crop is grown.
- In this disease, the inflorescence gets converted into green leafy ears.
- The causal organism of green ear disease of bajra is a fungus named **Sclerospora graminicola**. It is an obligate parasite.

Sugarcane Leafhopper :

- The Sugarcane Leafhopper (**Pyrilla perpusilla**) insect is a serious pest of sugarcane in Northern India.
- The pest is found gregariously on the under surface of the leaves where it sucks up plant sap that causes yellowing and eventually drying of leaves.
- Under low infestation yellow patches appear on the leaves.
- Photosynthesis is reduced resulting in the reduction of sucrose content of the juice by up to 30%.
- Leafhoppers secrete a sweet substance called honey dew that coats the leaves and attracts a blackish fungus, which reduces photosynthesis resulting in yield loss.

White Rust :

- White Rust is a disease in plants caused by the oomycetes **Albugo candida**.
- Plants susceptible to the disease generally include members of the family **Brassica** as mustard, broccoli, cauliflower, cabbage, etc.
- Signs and symptoms of infection include **chlorosis** on leaf surfaces, white blister-like growths on the underside of leaves and on the stems of the plant, and swelling of the roots.
- Fungicides are used for the treatment of white rust of crucifers.

Kernal Bunt :

- Kernal Bunt, also known as **partial bunt**, is a fungal disease of wheat, durum wheat, and triticale.
- It is caused by smut fungus **Tilletia indica**.
- It invades the kernels and obtains nutrients from the endosperm, leaving behind waste products with a disagreeable odour that makes bunted kernels too unpalatable for use in flour etc.
- Kernal bunt affects the quality of wheat.

Ergot disease :

- Ergot of cereals and grasses is a plant disease caused by the fungus **Claviceps purpurea**, which infects the developing grains of cereals and grasses.
- Ergot symptoms become evident during kernel formation when ergot bodies are formed in place of kernels.
- Ergot affects the productivity and quality of the crop and if livestock eat infected grain or hay it may cause a disease called **ergotism**.

Tungro disease :

- Tungro disease in rice is caused by the combination of two viruses, which are transmitted by leafhoppers. Viruses are (1) RTSV-rice tungro spherical virus and (2) RTBV-rice tungro bacilliform virus.
- It causes leaf discoloration, stunted growth, reduced tiller numbers and sterile or partly filled grains.
- It is one of the most destructive diseases of rice in South and Southeast Asia.

List of some diseases in Plants Caused by Viruses :

- | | |
|-------------------------|----------------------------|
| (1) Tobacco mosaic | (7) Lettuce mosaic |
| (2) Cucumber mosaic | (8) Maize mosaic |
| (3) Barley Yellow Dwarf | (9) Peanut stunt |
| (4) Bud Blight | (10) Yellow vein mosaic of |
| (5) Sugarcane mosaic | Lady's finger (okra) - Its |
| (6) Cauliflower mosaic | vector is Bemisia tabaci |
| | (white flies) |

Plant Diseases caused by Bacteria

Disease	Bacterium name
Potato Wilt	Pseudomonas solanacearum
Crown gall disease	Agrobacterium tumefaciens
Black rot of cabbage	Xanthomonas campestris
Canker of Citrus	Xanthomonas axonopodis

Plant Diseases caused by Fungus

Name of Disease	Fungus Name
Black rust of wheat	Puccinia graminis tritici
Loose smut of wheat	Ustilago nuda tritici
Early blight of potato	Alternaria solani
Late blight of potato	Phytophthora infestans
Wart disease of potato	Synchytrium endobioticum
Tikka disease of groundnut	Cercospora personata
Red rot of sugarcane	Colletotrichum falcatum
Brown leaf spot of rice	Sphaerulina oryzae
Wilt of arhar	Fusarium oxysporum

Plant Diseases due to lack of Nutrient

Name of deficiency	Disease / Symptom
Zinc (Zn)	Khaira disease of Paddy
Zinc (Zn)	White bud disease in maize
Boron (B)	Internal cork in apple fruit
Boron (B)	Browning in cauliflower
Boron (B)	Sugarbeet heart rot
Molybdenum (Mo)	Whiptail disease of cabbage

Question Bank

1. Which of the following statements is/are correct?

Viruses can infect

1. Bacteria
2. Fungi
3. Plants

Select the correct answer using the code given below:

- (a) 1 and 2 only
- (b) 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

I.A.S. (Pre) 2016

Ans. (d)

A virus is a small infectious agent that replicates only inside the living cells of other organisms. It can infect all types of life forms, from animals and plants to microorganisms, including bacteria and fungi.

2. Match List-I with List-II and select the correct answer from the code given below the lists.

List-I (Plant disease)	List-II (Factor)
A. Citrus Canker	1. Insect
B. Red rot disease of sugarcane	2. Deficiency of oxygen
C. Krishnakant disease of potato	3. Bacteria
D. Sahu disease of wheat	4. Fungus

Code :

A	B	C	D
(a) 1	3	4	2
(b) 1	2	3	4
(c) 4	1	3	2
(d) 3	4	2	1

U.P. P.C.S. (Pre) 2022

Ans. (*)

Citrus canker is a disease affecting Citrus species caused by the bacterium *Xanthomonas* (*X. axonopodis*; *X. campestris*). Infection causes lesions on the leaves, stems, and fruit of citrus trees, including lime, oranges, and grapefruit. This disease significantly affects the vitality of citrus trees, causing leaves and fruit to drop prematurely.

Red rot disease of sugarcane is characterized by interrupted red and white patches within the cane along with a sour alcoholic odour when the cane is split open. Caused by the fungus *Colletotrichum falcatum* (*Glomerella tucumanensis*), red rot first attracts attention by a yellowing and withering of the leaf, and eventually the entire plant dies.

Black heart (Krishnakant) disease of potatoes occurs primarily in storage when the tubers do not receive enough oxygen. Blackening of the tuber centre follows acute oxygen deficiency associated with either low temperature in confined storage or high field soil temperatures. The tissue dies from the inside out and turns jet black.

The ear cockle disease of wheat occurs in most wheat-growing parts of the world. In India, the disease commonly occurs in U.P., Punjab and Western part of Bihar, and is popularly known as 'sehun' disease. Nematodes are the causal organism of ear cockle disease. It is not caused by the insects.

Note : 'Sahu' disease of wheat is given in the question, which should be 'sehun'.

3. Which one of the following is not correctly matched ?

- (a) Wheat – Phalaris minor
- (b) Potato – Late blight
- (c) Bajra – Ergot
- (d) Sugarcane – Powdery mildew

U.P.P.C.S. (Mains) 2009

Ans. (d)

Powdery mildew is related with wheat, grape, barley, onion, apple and pea etc. not with sugarcane. Other pairs are correctly matched.

4. Which one of the following is not correctly matched?

Crop	Disease
(a) Rice	– Green hair
(b) Bajra	– Ergot
(c) Pea	– Powdery mildew
(d) Gram	– Wilt

U.P. Lower Sub. (Pre) 2015

Ans. (a)

Green hair disease is found in bajra. The symptom of green hair disease is the hairs of bajra becomes twisted. Green leaves and the hairs of bajra seems like a broom, as a result the plants remains dwarf. Rest of the options are correctly matched.

5. Which one of the following is not correctly matched?

Crop	Disease
(a) Paddy	– Wilt
(b) Mustard	– White rust
(c) Bajra	– Smut
(d) Groundnut	– Tikka

U.P.P.C.S. (Mains) 2017

Ans. (a)

Wilting occurs when leaves and other delicate aerial components of plant lose their turgidity, producing drooping, folding and rolling. It is mainly a disease of gram, chick pea, sugarcane, toor, etc. It is not a disease of paddy. Other pairs are correctly matched.

6. Which element deficiency causes chlorosis in maize plant?

- (a) Nitrogen (b) Zinc
(c) Copper (d) Manganese

U.P. U.D.A./L.D.A. (Pre) 2006

Ans. (b)

'Zn' (Zinc) is a micronutrient element which is required in little quantity by plants. Due to deficiency of Zn, chlorosis occurs in maize plants in which leaves of plants are undeveloped and plants become dwarf. The tip of maize leaves get white and they are mottled and yellow because of Zn deficiency.

7. White tip of maize leaves indicates the deficiency of

- (a) Fe (b) Mn
(c) N (d) Zn

U.P.P.C.S. (Mains) 2010

Ans. (d)

See the explanation of above question.

8. White Bud disease in plant is caused by deficiency of

- (a) Zinc (b) Copper
(c) Boron (d) Manganese

Jharkhand P.C.S. (Pre) 2016

Ans. (a)

Name given to Zinc (Zn) deficiency include khaira disease of rice, white bud of maize, mottle leaf (little leaf) or frenching of citrus and cotton.

9. Pyrilla is pest of which crop?

- (a) Wheat (b) Rice
(c) Sugarcane (d) Soyabean

M.P. P.C.S. (Pre) 1990

Ans. (c)

Pyrilla is leafhopper which sucks up leaves and plant sap of sugarcane. Under low infestation, yellow patches appear on the leaves that cause yellowing and eventually drying of leaves. That's why sugarcane crop growth disrupted. This insect is very harmful to sugarcane.

10. The deadly disease of groundnut is :

- (a) Powdery mildew (b) None of these
(c) Tikka (d) Damping off

Jharkhand P.C.S. (Pre) 2021

Ans. (c)

Tikka is a major disease affecting groundnut (and all its variety) in India. It occurs as well in almost all groundnut-growing countries of the world. It occurs as two distinct types of leaf spot, namely, early leaf spot and late leaf spot. These are caused by pathogenic fungus *Cercospora personata* and *Cercospora arachidicola*.

11. White rust is prominent fungal disease of –

- (a) Wheat (b) Mustard
(c) Rice (d) Millet

R.A.S./R.T.S. (Pre) 1996

Ans. (b)

White rust is a prominent fungal disease of mustard which spread by fungi *Albugo candida*. White lesions/patches appear on the stem, leaves and inflorescence of the plant, as a result the capacity of plant reduced. Fungicide should be used for the treatment of white rust.

12. Rice blast disease is a ____ disease.

- (a) Fungal (b) Bacterial
(c) Nutritional (d) All of the above

Chhattisgarh P.C.S. (Pre) 2024

Ans. (a)

The rice blast disease is caused by an ascomycetes fungus named *Piricularia oryzae* or *Magnaporthe oryzae* (*Magnaporthe grisea*). It can affect all above ground parts of a rice plant : leaf, collar, node, neck, parts of panicle, and sometimes leaf sheath. It is one of the most destructive diseases of rice. Rice blast causes economically significant crop loss annually worldwide.

13. Which among the following statements is not true?

- (a) Rust is a disease found in wheat.
(b) Early blight is a disease found in potato.
(c) False smut is a disease found in barley.
(d) Khaira is a disease found in paddy.

U.P. R.O./A.R.O. (Pre) 2017

Ans. (c)

False smut is a disease found in paddy (not in barley). It causes chalkiness of grains which leads to reduction in grain weight. It also reduces seed germination. Statements of other three options are correct.

14. Two common fungal diseases of wheat are :

- (a) Black rust & smut (b) White rust & smut
(c) Smut & leaf curl (d) Black rust & white rust

R.A.S./R.T.S. (Pre) 1996

Ans. (a)

Two common fungal diseases of the wheat crop are black rust and loose smut. Black rust of wheat is caused by *Puccinia graminis tritici* and loose smut of wheat is caused by *Ustilago nuda tritici*.